Q3: Engineer a Device-Status Register with Bitwise Operators

CODE: #include <stdio.h>

```
enum { LIGHT = 0, FAN = 1, AC = 2, HEATER = 3, PUMP = 4 };

unsigned char switch_on(unsigned char reg, int dev)
{
   return reg | (1u << dev);
}

unsigned char switch_off(unsigned char reg, int dev)
{
   return reg & ~(1u << dev);
}

unsigned char toggle(unsigned char reg, int dev)
{
   return reg ^ (1u << dev);
}

int is_on(unsigned char reg, int dev)
{
   return (reg >> dev) & 1u;
}

void show(unsigned char reg)
{
```

```
    for (int bit = 7; bit >= 0; bit--)
        putchar((reg >> bit) & 1u ? '1' : '0');

    putchar('\n');
}

int count_on(unsigned char reg)
{
    int count = 0;

    for (int bit = 0; bit < 8; bit++)
        count += (reg >> bit) & 1u;

    return count;
}

int main()
{
    unsigned char reg = 0;

    reg = switch_on(reg, LIGHT);
    reg = switch_on(reg, AC);
    show(reg);

    reg = toggle(reg, FAN);
    reg = switch_off(reg, LIGHT);
    show(reg);
```

```
    printf("AC is %s, PUMP is %s\n",
        is_on(reg, AC) ? "ON" : "OFF",
        is_on(reg, PUMP) ? "ON" : "OFF");

    printf("Devices ON: %d\n", count_on(reg));

    return 0;
}
```

